PIPING AND INSTRUMENTATION DIAGRAM FOR FIRE PROTECTION SYSTEM

NOTES :

DRAWING REFERENCE	DESCRIPTION	REV	DATE	DESCRIPTION	PREPR'D	CHECK'D	APRV'D	CHECKED	APPROVED
		A	16-12-2025	ISSUE FOR REVIEW	AI	AS	RTN		

PT. PERTAMINA EP APPROVAL

PROCET :

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP)
TANJUNG MIRING BARAT & KARANGAN TIMUR

OWNER :

skkmigas PERTAMINA EP

CONTRACTOR :

POG

SCALE :	DRAWING NO.	SHEET. NO.	REV
NTS	TMBP – SP001 – PS – PID – D101	01 OF 03	A

DRAWING TITLE :

PIPING AND INSTRUMENTATION DIAGRAM FOR FIRE PROTECTION SYSTEM

PN-003
FIRE WATER POND

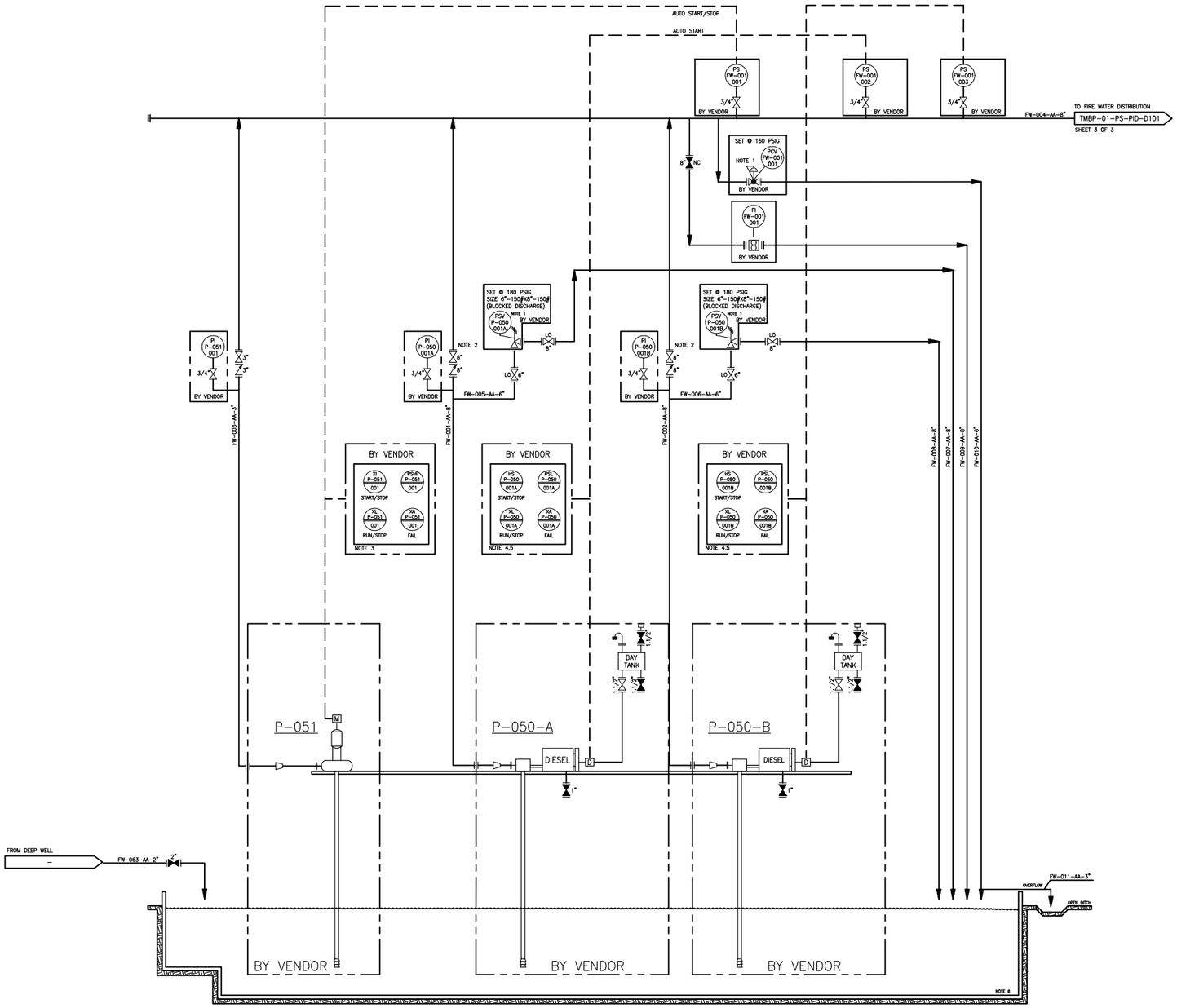
SIZE : 56 ft L x 49 ft W x 16 ft D
CAPASITY : 300000 GALLON
OPER. PRESS/ TEMP : ATM/AMB

P-050-A/B
VERTICAL FIRE WATER PUMP

CAPASITY : 1250 GPM
DISCH. PRESS : 157.74 PSIG
DIFF. HEAD : 388.35 ft
HHP : 122.44 HP

P-051
JOCKEY PUMP

CAPASITY : 15 GPM
DISCH. PRESS : 159.84 PSIG
DIFF. HEAD : 369.85 ft
HHP : 1.4 HP



PN-003

NOTES :
1. SET PRESSURE OF CIRCULATION RELIEF VALVE IS BELOW SHUT OFF PRESSURE PUMP
2. OSY GATE VALVE ATAU INDICATING BUTTERFLY VALVE
3. AUTO START JOCKEY PUMP AT 145 PSIG
   AUTO STOP JOCKEY PUMP AT 155 PSIG
4. AUTO START MAIN FIRE WATER PUMP AT 140 PSIG
   SECONDARY FIRE WATER PUMP AT 130 PSIG
5. BOTH MAIN AND SECONDARY FIRE WATER PUMP STOP FEATURE SHALL BE LOCAL ONLY
6. FIRE WATER POND INITIAL FILLING USING WATER TRUCK.
   SERVICE WATER FOR MAKE UP ONLY.

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H-001/002/003/004/005/006/007/008/009/010
FIRE HYDRANT

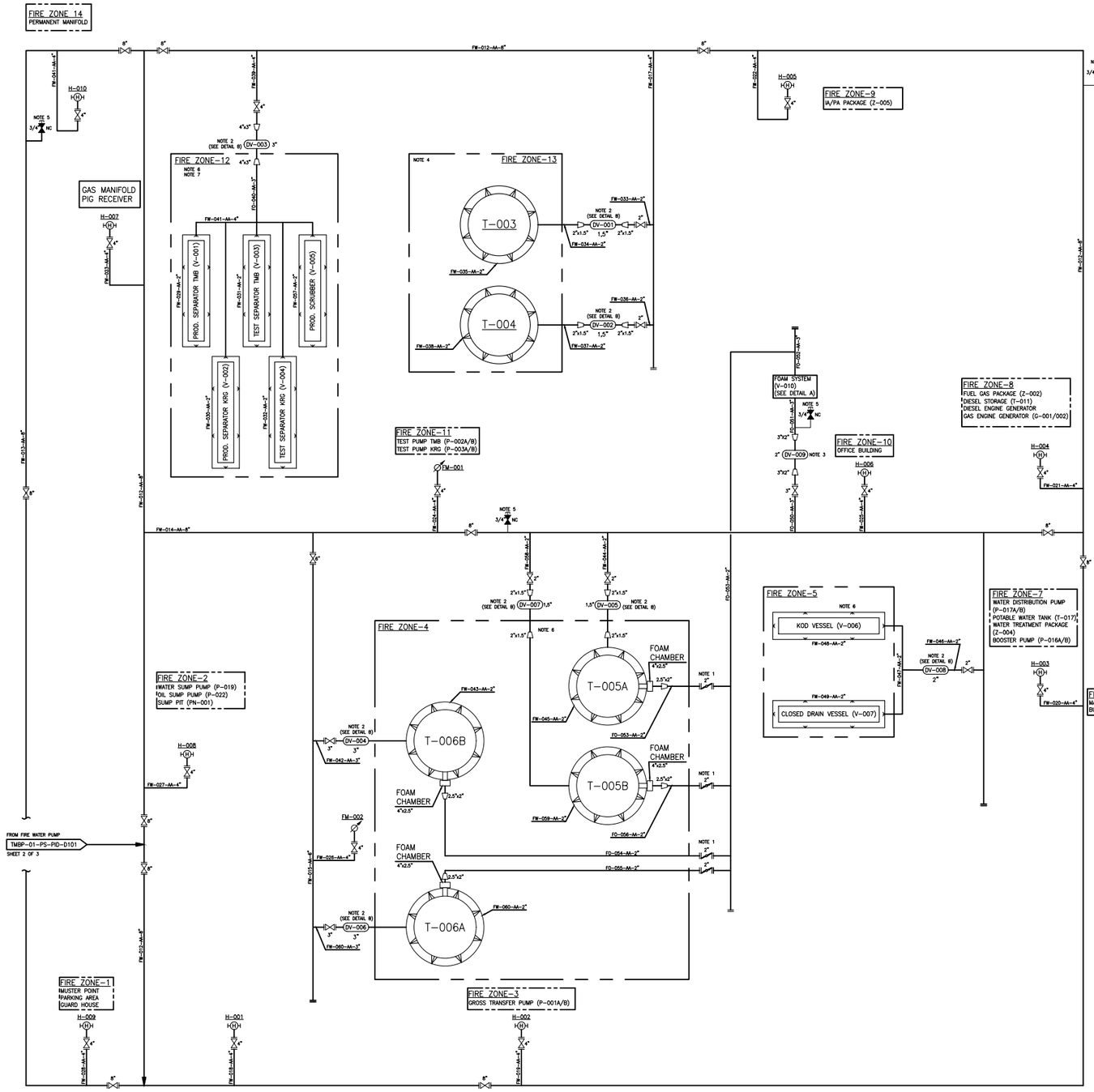
CAPASITY : 250 GPM
OPER. PRESS/ TEMP : 100 PSIG/AMB
MATERIAL : CARBON STEEL (VTA)

FM-001/002
FIRE MONITOR

CAPASITY : 500 GPM
OPER.PRESS/ TEMP : 100 PSIG/AMB
MATERIAL : BRONZE/BRASS (VTA)

V-010
FOAM BLADDER TANK

CAPASITY : 200 GALLON
DES. PRESS/ TEMP : VTA/VTA
OPER.PRESS/ TEMP : 100 PSIG/AMB
MATERIAL : (VTA)



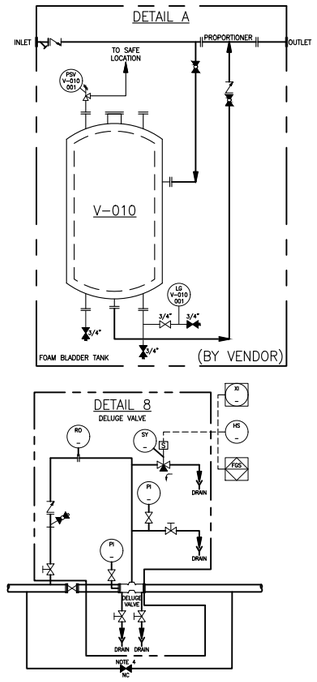


TABLE A
SPRAY NOZZLE

NO	TAG NO.	K FACTOR	SPRAY ANGLE	FIXED ANGLE	NOZZLE QUANTITY RING 1 (ROOF/TOP)	NOZZLE QUANTITY RING 2 (SHELL)
1	T-003	3	110°	90°	–	5
2	T-004	3	110°	90°	–	6
3	T-005A	3	110°	90°	–	5
4	T-005B	3	110°	90°	–	5
5	T-006A	3	110°	90°	–	10
6	T-006B	3	110°	90°	–	10
7	V-001	1.8	110°	45°	10	–
8	V-002	1.8	110°	45°	12	–
9	V-003	1.8	110°	45°	2	–
10	V-004	1.8	110°	45°	2	–
11	V-005	1.8	110°	45°	–	4
12	V-006	1.8	110°	45°	6	–
13	V-007	1.8	110°	45°	6	–

NOTES :
1. ALL BUTTERFLY VALVE FOR FOAM SOLUTION OPENING TO BE LOCATED TOGETHER IN SAFE AREA AND NEAR OPERATOR ACCESS.
2. DELUGE VALVE TO BE LOCATED NEAR ITS DESTINATION FIRE ZONE AND NEAR OPERATOR ACCESS.
3. DELUGE VALVE FOR FOAM SYSTEM TO BE LOCATED NEAR FOAM BLADDER TANK.
4. BYPASS LINE AND VALVE SIZE IS SAME WITH DELUGE VALVE SIZE.
5. CONNECTION FOR PRESSURE GAUGE/INDICATOR.
6. SPRAY NOZZLE QUANTITY FOR EACH EQUIPMENT INDICATED AT TABLE A.
7. THE 2 IN. OF LINE SIZE IS FOR SPRAY RING AT EACH SEPARATOR ONLY THE LINE FOR SPRAY RISER AND DELUGE VALVE IS 3 IN.

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